Project Management PW4 fs881, bta26, ac3847

Project Title: Home Audio System

Project Aims: The project aims to design, simulate, build, and verify a complete home audio system capable of processing and amplifying an audio signal from input to earphone output. The system integrates three interdependent subsystems: a 3-band graphic equalizer that can independently boost or cut bass, midrange, and treble frequency bands by up to 12 dB; a Class D switching amplifier that efficiently converts the equalised signal into a high-power output through pulse-width modulation and switching; and an OLED display providing a real-time spectrogram visualisation of the audio signal. The project applies a structured engineering workflow encompassing circuit design, simulation, breadboard implementation, and systematic bench verification at each stage prior to full system integration.

Project Scope: The scope of this project encompasses the design, construction, and verification of three electronic subsystems built across two breadboards and driven by an Arduino microcontroller. This includes the implementation of active bandpass filters forming the equalizer, a PWM modulator and half-bridge switching output stage forming the Class D amplifier, an RLC output filter matched to the load impedance, and embedded firmware for the OLED spectrogram. Each subsystem is individually verified using bench instrumentation before integration into the complete system. The project is limited to the breadboard prototype and does not extend to PCB design, enclosure fabrication, or multi-channel audio output.

Project SMART Objectives:

O1 - Graphic Equalizer Design and Simulation Design and simulate three MFB bandpass filters covering 100 Hz to 8 kHz in LTspice, achieving 0 dB flat response at neutral, 12 dB maximum gain per band, and ripple below 4 dB. All component values pre-calculated using E12 standard resistors. Achievable using available lab components and LTspice. Completed by end of Day 1.

O2 - Graphic Equalizer Build and Verification Build the full 3-band equalizer on a breadboard and verify using the function generator and oscilloscope. Measured BPF centre frequencies must be within 15% of target, ripple below 4 dB, and 12 dB gain confirmed per band. Achievable within one day using pre-simulated values. Completed by end of Day 2.

O3 - OLED Welcome Page Implement the OLED welcome page displaying the project name and team member names, confirmed by correct initialisation with no I2C errors and a 4-second transition to the spectrogram. Achievable within two hours using the Adafruit library on the Arduino Nano. Completed by end of Day 2.

O4 - Class D Modulator and Switching Stage Build the triangle wave integrator, LM393 comparator, and half-bridge switching stage with dead time circuit. Oscilloscope must confirm 100 kHz triangle wave, correct PWM output, and dead time of at least 0.5 µs on both transitions with no shoot-through. Achievable in one focused session. Completed by end of Day 3.

O5 - RLC Low-Pass Filter and Class D Output Measure earphone impedance with the DMM, build the matched RLC filter, and verify output ripple below 100 mV and DC offset below 0.4 V using the oscilloscope. Achievable within one afternoon using pre-calculated L and C lookup values. Completed by end of Day 4.

O6 - Full System Integration and Design Validation Connect all three subsystems and pass an audio signal through the complete chain to earphones, with the OLED spectrogram responding to potentiometer adjustment. Validated by a short recorded video. Achievable within the final session given all prior subsystems verified. Completed by end of Day 5.

Work Package 1: Project Coordination and QMS — Day 1

Summary: Define roles, establish documentation structure, and maintain project traceability throughout the week.

Tasks:

- T1.1 Define roles and assign lead responsibilities for circuit design, embedded programming, and bench testing
- T1.2 Build shared folder structure with subfolders for photos, measurements, code, and evidence
- T1.3 Draft and agree a file naming convention before any files are saved
- T1.4 Create and maintain a file index covering all project files and their associated milestone
- T1.5 Brief daily check-in each morning to review progress and reprioritise tasks if any subsystem is behind

Work Package 2: Graphic Equalizer — Day 2

Summary: Design, build, and verify the 3-band graphic equalizer and OLED welcome page.

Tasks:

- T2.1 Select component values for all three MFB bandpass filters and verify designs meet frequency and gain targets (M2a)
- T2.2 Build BPF2 midrange first on the breadboard, test with function generator, then build BPF1 and BPF3 (M2c)
- T2.3 Wire the full equalizer topology with summing amplifier and verify neutral response, max gain, and ripple with oscilloscope (M2c)
- T2.4 Implement and verify the OLED welcome page displaying project name and team member names on the Arduino Nano (M2d)

Work Package 3: Class D Amplifier — Day 3

Summary: Build and verify the triangle wave generator, PWM modulator, and half-bridge switching output stage.

Tasks:

- T3.1 Build the triangle wave integrator circuit and verify the 100 kHz waveform and amplitude on the oscilloscope (M3a)
- T3.2 Build the LM393 comparator and verify the PWM output correctly modulates a 1 kHz test sine wave (M3a)
- T3.3 Build the NPN gate driver and half-bridge switching stage with RC dead time circuit (M3b)
- T3.4 Measure dead time on the oscilloscope and confirm it is equal on both switching transitions with no shoot-through (M3b)

Work Package 4: RLC Filter, Spectrogram, and Integration — Days 4 and 5

Summary: Complete the output filter, OLED spectrogram, and full system integration.

Tasks:

- T4.1 Measure earphone DC resistance with DMM, select appropriate inductor and capacitor, build and verify the RLC output filter (M4a)
- T4.2 Measure Class D amplifier output waveform and confirm ripple below 100 mV and DC offset below 0.4 V (M4b)
- T4.3 Build peak detector hardware for all three bands, wire to Arduino analog inputs, and complete spectrogram code (M4c)
- T4.4 Connect all three subsystems and connect earphones via DC blocking capacitor to verify the complete audio chain (M4d)

Work Package 5: Risk Management and Quality Assurance — Ongoing

Summary: Monitor and mitigate risks throughout the week and ensure all evidence meets quality standards before sign-off.

Tasks:

- T5.1 Enforce sequential verification: no subsystem is connected to the next until individually verified with bench instruments
- T5.2 Confirm all oscilloscope screenshots and photos are saved with correct filenames at the end of each session
- T5.3 Confirm DC offset is below 0.4 V before earphones are connected
- T5.4 Final QMS check on Day 5: verify file index is complete and all milestone evidence is saved and accessible

Project Deliverables

Deliverable ID	Deliverable	Work Package	Format
D1	Measured frequency response for each BPF individually and the full EQ at neutral, max gain, and full boost	WP2	Oscilloscope screenshots
D2	OLED welcome page photograph	WP2	JPEG
D3	Oscilloscope screenshots of triangle wave and PWM output	WP3	JPEG
D4	Dead time measurement with oscilloscope cursor labels on both transitions	WP3	Oscilloscope screenshot
D5	RLC filter frequency response measured with function generator and oscilloscope	WP4	Oscilloscope screenshots
D6	Class D output waveform, ripple measurement, and DC offset measurement	WP4	Oscilloscope screenshots
D7	OLED spectrogram photograph showing three bands responding to input signal	WP4	JPEG
D8	Full system verification evidence showing audio output from earphones and spectrogram responding to potentiometer adjustment	WP4	Oscilloscope screenshots and photographs
D9	Completed file index and QMS folder with all evidence organised by milestone	WP5	Shared folder

Project Milestones

Milestone	Description	Work Package	Target
M1	Roles assigned, folder structure built, naming convention agreed, file index created	WP1	End of Day 1
M2a	Component values finalised for all three bandpass filters and gain targets confirmed	WP2	Day 2 AM
M2b	Full equalizer built and verified: neutral response, 12 dB gain, ripple below 4 dB confirmed on oscilloscope	WP2	Day 2 PM
M2c	OLED welcome page confirmed working with correct names and 4-second delay to spectrogram	WP2	End of Day 2
M3a	Triangle wave at 100 kHz and PWM output confirmed on oscilloscope	WP3	Day 3 AM
M3b	Dead time measured and equal on both transitions with no shoot-through current	WP3	Day 3 PM
M4a	RLC filter built and verified with ripple below 100 mV and DC offset below 0.4 V confirmed	WP4	Day 4 PM
M4b	Three-band OLED spectrogram smoothly tracking amplitude on all three channels	WP4	Day 5 AM
M4c	Full system integration complete with audio confirmed from earphones and spectrogram responding to potentiometer adjustment	WP4	End of Day 5
M5	File index complete, all evidence saved and accessible, QMS folder signed off	WP5	End of Day 5